

An Analysis of Semi-supervised learning

Semi-supervised learning

Semi-supervised learning -- Part 1

Laplacian regularization Laplacian regularization has been historically approached through graph-Laplacian. Graph-based methods for semi-supervised learning use a graph representation of the data, with a node for each labeled and unlabeled example. The graph may be constructed using domain knowledge or similarity of examples; two common methods are to connect each data point to its k nearest neighbors or to examples within some distance ϵ . The weight W_{ij} of an edge between x_i and x_j is then set to $e^{-\|x_i - x_j\|^2 / \epsilon^2}$. Within the framework of manifold regularization, the graph serves as a proxy for the manifold. A term is added to the standard Tikhonov regularization problem to enforce smoothness of the solution relative to the manifold (in the intrinsic space of the problem) as well as relative to the ambient input space. The minimization problem becomes

Semi-supervised learning -- Part 2

Cluster assumption The data tend to form discrete clusters, and points in the same cluster are more likely to share a label (although data that shares a label may spread across multiple clusters). This is a special case of the smoothness assumption and gives rise to feature learning with clustering algorithms.

Semi-supervised learning -- Part 3

An exact solution is intractable due to the non-convex term $(1 - |f(x)|) +$, so research focuses on useful approximations. Other approaches that implement low-density separation include Gaussian process models, information regularization, and entropy minimization (of which TSVM is a special case).

Semi-supervised learning -- Part 4

$f^* = \underset{f}{\operatorname{argmin}} \left(\sum_{i=1}^l (1 - y_i f(x_i)) + \lambda \sum_{i=1}^l |f(x_i)| + \lambda \sum_{i=1}^l |f(x_{l+1+u})| \right)$

Semi-supervised learning -- Part 5

Heuristic approaches Some methods for semi-supervised learning are not intrinsically geared to learning from both unlabeled and

labeled data, but instead make use of unlabeled data within a supervised learning framework. For instance, the labeled and unlabeled examples $x_1, \dots, x_{l+u}$ may inform a choice of representation, distance metric, or kernel for the data in an unsupervised first step. Then supervised learning proceeds from only the labeled examples. In this vein, some methods learn a low-dimensional representation using the supervised data and then apply either low-density separation or graph-based methods to the learned representation. Iteratively refining the representation and then performing semi-supervised learning on said representation may further improve performance. Self-training is a wrapper method for semi-supervised learning. First a supervised learning algorithm is trained based on the labeled data only. This classifier is then applied to the unlabeled data to generate more labeled examples as input for the supervised learning algorithm. Generally only the labels the classifier is most confident in are added at each step. In natural language processing, a common self-training algorithm is the Yarowsky algorithm for problems like word sense disambiguation, accent restoration, and spelling correction. Co-training is an extension of self-training in which multiple classifiers are trained on different (ideally disjoint) sets of features and generate labeled examples for one another.

Semi-supervised learning -- Part 6

$$T(L, f) = \sum_{i,j=1}^{l+u} W_{ij} (f_i - f_j)^2 \approx \int M \nabla M f(x) d p(x)$$
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 The graph-based approach to Laplacian regularization is to put in relation with finite difference method. The Laplacian can also be used to extend the supervised learning algorithms: regularized least squares and support vector machines (SVM) to semi-supervised versions Laplacian regularized least squares and Laplacian SVM.

Semi-supervised learning -- Part 7

Generative models Generative approaches to statistical learning first seek to estimate $p(x|y)$, the distribution of data points belonging to each class. The probability $p(y|x)$ that a given point x has label y is then proportional to $p(x|y)p(y)$ by Bayes' rule. Semi-supervised learning with generative models can be viewed either as an extension of supervised learning (classification plus information about $p(x)$) or as an extension of unsupervised learning (clustering plus

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some labels). Generative models assume that the distributions take some particular form $p(x|y, \theta)$ parameterized by the vector θ . If these assumptions are incorrect, the unlabeled data may actually decrease the accuracy of the solution relative to what would have been obtained from labeled data alone. However, if the assumptions are correct, then the unlabeled data necessarily improves performance. The unlabeled data are distributed according to a mixture of individual-class distributions. In order to learn the mixture distribution from the unlabeled data, it must be identifiable, that is, different parameters must yield different summed distributions. Gaussian mixture distributions are identifiable and commonly used for generative models. The parameterized joint distribution can be written as $p(x, y|\theta) = p(y|\theta)p(x|y, \theta)$ by using the chain rule. Each parameter vector θ is associated with a decision function $f_\theta(x) = \arg\max_y p(y|x, \theta)$. The parameter is then chosen based on fit to both the labeled and unlabeled data, weighted by λ :

Semi-supervised learning -- Part 8

More formally, semi-supervised learning assumes a set of l independently identically distributed examples $x_1, \dots, x_l \in X$ with corresponding labels $y_1, \dots, y_l \in Y$ and u unlabeled examples $x_{l+1}, \dots, x_{l+u} \in X$ are processed. Semi-supervised learning combines this information to surpass the classification performance that can be obtained either by discarding the unlabeled data and doing supervised learning or by discarding the labels and doing unsupervised learning. Semi-supervised learning may refer to either transductive learning or inductive learning. The goal of transductive learning is to infer the correct labels for the given unlabeled data $x_{l+1}, \dots, x_{l+u}$ only. The goal of inductive learning is to infer the correct mapping from X to Y . It is unnecessary (and, according to Vapnik's principle, imprudent) to perform transductive learning by way of inferring a classification rule over the entire input space; however, in practice, algorithms formally designed for transduction or induction are often used interchangeably.